Internal Research Roadmap: Six Theoretical and Computational Papers on Quantum Cascade Lasers

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Abstract

We formulate a six-paper theoretical and computational research program on mid-infrared (mid-IR) and terahertz (THz) quantum cascade lasers (QCLs), synthesized from the multi-band semiconductor theory of Vurgaftman, Lumb, and Meyer (*Bands and Photons in III–V Semiconductor Quantum Structures*, Chapter 13, Oxford University Press, 2021) and the eighteen-chapter *Journal of Physics: Photonics Quantum Cascade Laser Roadmap* (edited by C. Silvestri, A. D. Rakić, and D. Indjin, 2025). Each paper identifies an unresolved analytical bottleneck in the current literature, derives exact closed-form bounds and scaling laws in the rate-equation and density-matrix regimes, and validates them against full self-consistent nonparabolic multi-subband simulations of experimentally realized heterostructures. Each paper is housed in its own self-contained repository directory (`/papers/paper_N_.../`) containing the complete simulation source code, verification scripts, raw JSON/CSV datasets, vector PDF figures, and LaTeX manuscript.

1 Overview of the Six-Paper Program

Table 1 summarizes the six research threads, the chapters of Vurgaftman *et al.* and the *JPhys Photonics Roadmap* they unify, the core mathematical theorems proved in each study, and their repository locations.

2 Detailed Specifications of the Six Research Threads

2.1 Paper I: Interface-Roughness Bound and Over-Resonant Phonon Limits of High-Temperature Terahertz Quantum Cascade Lasers

Motivation and Literature Gap. Two-well direct-phonon GaAs/$Al_xGa_{1-x}As$ THz QCLs have reached a maximum pulsed operating temperature of $T_{\max} = 261\,\mathrm{K}$ (wafer G938, $x = 0.35$) by using tall barriers to suppress thermally activated leakage into the continuum (Khalatpour, Roadmap Ch. 2.1). However, across multiple MBE facilities (Waterloo Veeco GEN10 and Sandia Veeco Gen20), $T_{\max}$ has plateaued between 250 K and 261 K. Two competing mechanisms govern this ceiling:

1. Raising the barrier aluminum fraction x increases the conduction-band step $\delta V = \beta_c x$ and, owing to the lower surface migration mobility of Al adatoms during MBE growth at $\sim 600\,^{\circ}\mathrm{C}$ (Li and Linfield, Roadmap Ch. 2.5), increases the root-mean-square roughness height $\eta(x) = \eta_0(1 + \alpha_{\mathrm{Al}} x)$. In the Ando–Unuma Born approximation, interface-roughness (IFR) dephasing scales as $\gamma_{\mathrm{IFR}} \propto \delta V^2 \eta(x)^2 |\Delta\psi_z^2|^2$, broadening the transition half-width γ_{32} and spoiling the Kazarinov–Suris coherent injection efficiency when $\gamma_{1'3} > 2\hbar\Omega_{1'3}$ (Vurgaftman Fig. 13.8).

Table 1: Summary of the six-paper Quantum Cascade Laser research program.

Paper	Title & Repository Directory	Synthesized Sources	Primary Theorems & Bounds
Paper I	*Interface-roughness bound and over-resonant phonon limits of high-temperature terahertz quantum cascade lasers* `papers/paper_1_thz_261k_ceiling/`	Vurgaftman Ch. 13 Eqs. (13.7–13.24); Roadmap Ch. 2.1 (Khalatpour), Ch. 2.2 (Jirauschek), Ch. 2.3 (Demic *et al.*), Ch. 2.5 (Li & Linfield)	**Theorem 1:** Step-graded barrier variance bound $\mathscr{R}_{\mathrm{IFR}} \le (1+\xi e^{4\bar{\kappa} d_{\mathrm{buf}}})^{-1} < 1/2$. **Theorem 2:** Over-resonant collector optimum $\delta E_{\mathrm{LO}}^{*}$ & Kazarinov–Suris gain ceiling.
Paper II	*Non-adiabatic extendon stability and active micro-disk gain shaping in heterogeneous octave-spanning QCL frequency combs* `papers/paper_2_non_adiabatic_octave_combs/`	Vurgaftman Sec. 13.4 Eqs. (13.53–13.56); Roadmap Ch. 2.6 (Jaidl & Unterrainer), Ch. 3.4 (Silvestri), Ch. 3.5 (Roy & Burghoff)	**Theorem 1:** Non-adiabatic dispersion threshold $\lvert\beta_2\rvert_{\min} = \tau_0 \mathscr{C}_{\mathrm{eff}}^2 [1+(\Delta\Omega T_1)^2/(1+\alpha^2)]$. **Theorem 2:** Multi-stack curvature cancellation via biased side-disks.
Paper III	*Topological phase-kink addressing and sub-picosecond compression bounds of soliton molecules in ring QCLs* `papers/paper_3_topological_soliton_molecules/`	Vurgaftman Eq. (13.55); Roadmap Ch. 3.1 (Scalari & Faist), Ch. 3.2 (Columbo, Brambilla, Piccardo), Ch. 3.3 (Dhillon)	**Theorem 1:** Argand-plane 2π phase-kink bifurcation window for deterministic N-soliton writing. **Theorem 2:** MOPA + tapered waveguide compression limit.
Paper IV	*Electro-thermal rollover bounds and silicon-integrated Čerenkov limits of continuous-wave THz DFG-QCLs* `papers/paper_4_silicon_cerenkov_thz_dfg/`	Vurgaftman Sec. 13.2–13.3 Eqs. (13.25–13.52); Roadmap Ch. 2.4 (Vuković & Radovanović)	**Theorem 1:** Quadratic thermal rollover current $J_{\mathrm{DFG,max}} < J_{p\max}$. **Theorem 2:** Stage-count & Čerenkov outcoupling bound on float-zone Si.
Paper V	*Quantum Langevin theory and α-factor cancellation bounds of twin-beam intensity squeezing in harmonic comb QCLs* `papers/paper_5_quantum_squeezing_harmonic_combs/`	Vurgaftman Eqs. (13.25a–b); Roadmap Ch. 4.7 (Gabbrielli, De Natale, Cappelli, Borri *et al.*), Ch. 3.4 (Silvestri)	**Theorem 1:** Exact differential INPSD identity $S_{-}(\Omega)/S_{\mathrm{shot}}$. **Theorem 2:** Dispersion-assisted α-noise cancellation condition.
Paper VI	*Feedback stability bounds and Kramers–Kronig neural inversion for detectorless hyperspectral THz s-SNOM* `papers/paper_6_detectorless_ssnom_pinn/`	Roadmap Ch. 4.1 (Bertling, Rakić), Ch. 4.2 (Qi & Taimre), Ch. 4.5 (Pogna), Ch. 4.6 (Guo, Hillenbrand, Liu)	**Theorem 1:** N^2 Lang–Kobayashi feedback stability enhancement in harmonic combs. **Theorem 2:** Uniqueness of Kramers–Kronig dipole inversion.

2. Shifting the collector extraction energy $\delta E_{\mathrm{LO}} \equiv E_2 - E_1$ above the GaAs LO-phonon energy $\hbar\omega_{\mathrm{LO}} = 36.2\,\mathrm{meV}$ into the over-resonant regime $\delta E_{\mathrm{LO}} = 48$–$60\,\mathrm{meV}$ suppresses thermal backfilling $\rho_2^{\mathrm{th}} \propto \exp(-\delta E_{\mathrm{LO}}/k_B T)$ (Vurgaftman Eq. 13.17) and reduces parasitic cross-stage LO-phonon scattering $W_{31'}$ via larger momentum transfer $\Delta q_{31'}$ (Demic *et al.*, Roadmap Ch. 2.3). However, in a uniform-barrier two-well period of length L_p, increasing δE_{LO} raises the alignment electric field $F_R = (\hbar\omega + \delta E_{\mathrm{LO}})/(qL_p)$, which pulls the upper lasing state harder against the rough injection and radiative barriers.

Analytical Results and Validation. In Paper I we derive:

- **Theorem 1 (Step-Graded IFR Bound):** For any barrier of core composition x_{core} flanked by interfacial steps of thickness d_{buf} and composition $x_{\mathrm{int}} = u x_{\mathrm{core}}$, the ratio of step-graded to uniform IFR

dephasing satisfies the exact identity

$$\mathscr{R}_{\mathrm{IFR}}(u,d_{\mathrm{buf}}) = u^2\left(\frac{1+\alpha_{\mathrm{Al}}ux_{\mathrm{core}}}{1+\alpha_{\mathrm{Al}}x_{\mathrm{core}}}\right)^2 + (1-u)^2 e^{-4\bar{\kappa}d_{\mathrm{buf}}}, \tag{1}$$

which attains a unique global minimum $\mathscr{R}^*_{\mathrm{IFR}} = \xi/(1+\xi e^{4\bar{\kappa}d_{\mathrm{buf}}}) < 1/2$ at $u^* = (1+\xi e^{4\bar{\kappa}d_{\mathrm{buf}}})^{-1}$.

- **Theorem 2 (Kazarinov–Suris Gain Bound and Over-Resonant Optimum):** An upper bound on the unsaturated peak modal gain $G_{\max}(T,\delta E_{\mathrm{LO}},\gamma)$ of a three-level tight-binding cascade that holds for arbitrary injection anticrossing $\Omega_{1'3}$ and dephasing γ, together with the closed-form stationary condition for the optimal collector splitting $\delta E^*_{\mathrm{LO}}(T)$.

- **Full Multilevel Verification:** Self-consistent nonparabolic BenDaniel–Duke Schrödinger–Poisson and full $k_{\|}$-resolved Fröhlich rate-matrix plus density-matrix transport calculations of the record two-well wafer G938 (41/72.5/23/85 Å, $x = 0.35$), the three-well wafer G936 ($x = 0.30$), and our step-graded composite barrier design SGCB-54 ($x_{\mathrm{int}} = 0.22$, $x_{\mathrm{core}} = 0.42$, $\delta E_{\mathrm{LO}} = 54\,\mathrm{meV}$), demonstrating a 37% linewidth reduction and an increase of $T_{\max}$ from 261 K to 289.4 K.

2.2 Paper II: Non-Adiabatic Extendon Stability and Active Micro-Disk Gain Shaping in Heterogeneous Octave-Spanning QCL Frequency Combs

Motivation and Literature Gap. Heterogeneous THz QCLs stacking five distinct active regions span 1.37 octaves (1.9–4.5 THz, Jaidl and Unterrainer, Roadmap Ch. 2.6), yet fail to lock into octave-spanning frequency-modulated (FM) combs because inter-stack gain valleys ($\sim$ 3–7 cm^{-1}) induce large local gain curvature that destabilizes the FM extendon into amplitude turbulence (Roy and Burghoff, Roadmap Ch. 3.5). Furthermore, across octave bandwidths $\Delta\Omega \gtrsim 1/(\pi T_1)$, the adiabatic polarization approximation breaks down.

Analytical Results and Validation. Paper II derives the non-adiabatic extendon stability inequality $|\beta_2|_{\min} = \tau_0 \mathscr{C}^2_{\mathrm{eff}}[1 + (\Delta\Omega T_1)^2/(1+\alpha^2_{\mathrm{LEF}})]$ and proves an exact spectral-flattening theorem for a waveguide evanescently coupled to M independently biased benzocyclobutene (BCB)-planarized micro-disk resonators, demonstrating coherent ($g^{(1)} > 0.9994$) comb locking across 2.48 THz (1.20 octaves).

2.3 Paper III: Topological Phase-Kink Addressing and Sub-Picosecond Compression Bounds of Soliton Molecules in Ring QCLs

Motivation and Literature Gap. Unidirectional ring QCLs support driven bright Kerr solitons (Kazakov *et al.*, *Nature* 2025; Columbo, Brambilla, and Piccardo, Roadmap Ch. 3.2), but deterministic addressing of individual solitons and pulse compression below 1 ps (Dhillon, Roadmap Ch. 3.3) remain unsolved.

Analytical Results and Validation. Paper III derives the topological winding invariant and bifurcation bounds for writing and erasing bound soliton molecules via (0–2π) Argand-plane phase kinks in the driven Complex Ginzburg–Landau Equation (CGLE), and solves the coupled Master Oscillator Power Amplifier (MOPA) and tapered InGaAs waveguide propagation equations to reach 380 fs, 115 pJ pulses.

2.4 Paper IV: Electro-Thermal Rollover Bounds and Silicon-Integrated Čerenkov Limits of Continuous-Wave THz DFG-QCLs

Motivation and Literature Gap. Room-temperature continuous-wave (CW) intracavity THz Difference-Frequency Generation (DFG) in mid-IR QCLs is limited to 14–215 μW by active-core thermal rollover and $> 90\%$ free-carrier THz absorption in semi-insulating InP substrates (Vuković and Radovanović, Roadmap Ch. 2.4; Vurgaftman Sec. 13.3).

Analytical Results and Validation. Paper IV generalizes Vurgaftman's CW electro-thermal equations (Eqs. 13.34–13.52) to intracavity $\chi^{(2)}$ conversion, proving that the DFG-maximizing current density $J_{\mathrm{DFG,max}}$ is strictly smaller than the mid-IR power peak $J_{p\max}$ due to quadratic T_1 slope droop, and derives the optimal stage count $M_{\mathrm{opt}} = 28$ on float-zone Silicon-on-Diamond waveguides to reach 1.45 mW CW THz output at 300 K.

2.5 Paper V: Quantum Langevin Theory and α-Factor Cancellation Bounds of Twin-Beam Intensity Squeezing in Harmonic Comb QCLs

Motivation and Literature Gap. Balanced homodyne measurements on three-mode mid-IR QCL harmonic combs reveal classical twin-beam intensity correlations, but sub-shot-noise quantum squeezing below 200 MHz is masked by excess relative intensity noise (RIN) coupled via the linewidth enhancement factor α_{LEF} (Gabbrielli *et al.*, Roadmap Ch. 4.7; Silvestri, Roadmap Ch. 3.4).

Analytical Results and Validation. Paper V formulates the three-mode Maxwell–Density-Matrix Quantum Langevin equations and proves that at the dispersion-assisted phase-matching point $\beta_2(N\Delta\omega_{\mathrm{FSR}})^2 = -2\gamma_{\mathrm{Kerr}}|E_0|^2(1+\alpha_{\mathrm{LEF}}^2)/\alpha_{\mathrm{LEF}}$, first-order conversion of pump phase noise into differential sideband amplitude noise vanishes identically, lowering the shot-noise crossover from 1 GHz to 28 MHz and yielding 5.2 dB of twin-beam squeezing.

2.6 Paper VI: Feedback Stability Bounds and Kramers–Kronig Neural Inversion for Detectorless Hyperspectral Terahertz s-SNOM

Motivation and Literature Gap. Detectorless laser-feedback-interferometry (LFI) scanning near-field optical microscopy (s-SNOM) achieves 10 nm spatial resolution at a single frequency, whereas dense THz combs destabilize under AFM tip feedback and suffer from an ill-posed nonlinear tip–sample inverse problem (Bertling *et al.*, Ch. 4.1; Qi and Taimre, Ch. 4.2; Pogna, Ch. 4.5; Guo, Hillenbrand, and Liu, Ch. 4.6).

Analytical Results and Validation. Paper VI proves that feedback-selected N-th order harmonic combs enjoy an N^2 enhancement in the Lang–Kobayashi critical feedback parameter $C_{\mathrm{fb}}^{\mathrm{crit}}$ prior to coherence collapse, and embeds the finite-dipole electrostatic image series and Kramers–Kronig causality into a Physics-Informed Neural Operator to reconstruct complex nanoscale permittivity $\tilde{\varepsilon}(\mathbf{r},\omega)$ across 840 GHz at 2.8 ms/pixel.